INTERNSPARK PORTFOLIO REPORT: TASK 2

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Domain: Machine Learning Intern

Project Milestone: *Task 2 (Supervised Regression Workflow)*

1. Project Objective & Scope

The objective of this project was to engineer and evaluate an end-to-end supervised machine learning regression pipeline capable of predicting median house values across distinct California census tracts. Utilizing structural attributes (e.g., total rooms, bedrooms) and socioeconomic indicators (e.g., median income, geographic location), the pipeline benchmarks a baseline linear model against a robust non-linear ensemble method.

2. Exploratory Data Analysis (EDA) Insights

Statistical distribution analysis of the target variable (MedHouseVal) revealed a significant right-skewed distribution with a distinct cap at **$500,000** ($5.0$ in standardized units), signifying truncated outlier recording.

Feature interaction mapping via a Pearson Correlation Matrix identified a commanding positive linear correlation of **$+0.69$** between **Median Income (MedInc)** and the target valuation. In contrast, purely structural volume metrics (such as TotalRooms and TotalBedrooms) displayed minimal individual linear predictive power, highlighting the necessity for advanced non-linear modeling.

3. Pipeline Engineering & Preprocessing

To guarantee unbiased model evaluation, the dataset was systematically prepared using a strict modular workflow:

- **Data Partitioning:** An 80/20 train-test split was executed using a fixed pseudo-random seed (random_state=42) to isolate validation records and prevent data leakage.
- **Feature Normalization:** Because structural feature magnitudes (e.g., population counts) vastly exceeded geographic coordinates, a StandardScaler was fitted strictly to the training partition. This applied a standard Z-score transformation ($z = \frac{x - \mu}{\sigma}$), mapping all attributes to a uniform scale centered around zero to neutralize unit variance bias.

4. Quantitative Model Evaluation & Benchmarking

The processed feature space was modeled using two competing architectures: an **Ordinary Least Squares (OLS) Linear Regression** baseline and a highly optimized **Random Forest Regressor** ensemble constrained to a max_depth=15 to ensure efficient deployment size. Performance was evaluated using three industry-standard regression metrics:

Model Architecture	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)	R-squared Variance (R2)
Linear Regression Baseline	*0.5332*	*0.7456*	*0.5758*
Random Forest Regressor	*0.3274*	*0.5051*	*0.8053*

5. Analytical Conclusion

The empirical results demonstrate that the **Random Forest Regressor drastically outperformed** the baseline Linear Regression model across all monitored criteria. While the linear model was constrained by its rigid hyper-plane assumptions, the Random Forest architecture successfully captured complex, non-linear geographic dependencies (leveraging latitude and longitude interactions) and localized wealth density patterns.

The finalized ensemble model and corresponding scaler transformations were successfully serialized using joblib with an optimized compression coefficient (compress=3) to ensure production readiness within standard repository storage boundaries.